

Project Documentation

Credit Card Approval Prediction

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Milestone 1: Data Collection

Dataset Source

The project uses the publicly available Credit Card Approval Prediction dataset from Kaggle (CC0 license), published by **rikdifos**. The dataset contains real anonymized credit application records from a financial institution.

Dataset: rikdifos/credit-card-approval-prediction

Files

File	Rows	Columns	Description
application_record.csv	438,557	18	One row per applicant with demographic and financial attributes
credit_record.csv	1,048,575	3	Monthly credit behavior records per applicant (ID, MONTHS_BALANCE, STATUS)

Key Columns — application_record.csv

Column	Type	Description
ID	Integer	Unique applicant identifier (join key)
CODE_GENDER	Categorical	M/F
FLAG_OWN_CAR	Categorical	Y/N
FLAG_OWN_REALTY	Categorical	Y/N
CNT_CHILDREN	Integer	Number of children
AMT_INCOME_TOTAL	Float	Annual income

NAME_INCOME_TYPE	Categorical	Working, Pensioner, Commercial associate, etc.
NAME_EDUCATION_TYPE	Categorical	Higher education, Secondary, Academic degree, etc.
Column	Type	Description
NAME_FAMILY_STATUS	Categorical	Married, Single, Separated, etc.
NAME_HOUSING_TYPE	Categorical	House/apartment, Rented apartment, etc.
DAYS_BIRTH	Integer	Days since birth (negative — older → more negative)
DAYS_EMPLOYED	Integer	Days since employment start (365243 = unemployed)
OCCUPATION_TYPE	Categorical	Laborers, Core staff, Managers, etc.
CNT_FAM_MEMBERS	Float	Number of family members

Key Columns — credit_record.csv

Column	Description
ID	Applicant identifier (foreign key)
MONTHS_BALANCE	Months relative to current month (0 = current, -1 = last month)
STATUS	Credit behavior: 0=no debt, 1=30-59 DPD, 2=60-89 DPD, 3=90-119 DPD, 4=120-149 DPD, 5=150+ DPD, C=closed, X=no credit

Milestone 2: Exploratory Data Analysis (EDA)

Key Findings

- **Gender distribution:** 66.7% Female, 33.3% Male applicants
- **Car ownership:** 34.3% own a car; 69.8% own real estate
- **Income distribution:** Right-skewed; median ~\$1,57,500 (scaled from raw USD values)
- **Age distribution:** Peak applicants aged 35-50 years
- **Employment duration:** Wide distribution; 365243 sentinel value identifies unemployed applicants
- **Income type:** Working (52%) and Commercial associate (22%) are most common
- **Missing values:** OCCUPATION_TYPE has 31.2% missing values — imputed with "Unknown" • **Class distribution** (post-target engineering): ~87% Approved, ~13% Rejected

EDA Notebook

Full EDA with visualizations available in: `notebooks/Credit_Card_Approval_Analysis.ipynb`

Sections: Setup → Data Loading → EDA → Target Engineering → Feature Engineering → Preprocessing → Model Training → Model Comparison → Classification Reports → Best Model Selection

Milestone 3: Feature Engineering & Preprocessing

Target Variable Engineering

Credit behavior from `credit_record.csv` is mapped to a binary label:

- STATUS 1-5 (30+ days past due at any point) → **Rejected (0)**
- STATUS 0/C/X only (no late payments) → **Approved (1)**
- STATUS X only (no credit evaluated) → **Excluded**

This logic mirrors real-world bank risk assessment: any historical late payment flags the applicant as high-risk.

Derived Features

Feature	Formula	Rationale
AGE_YEARS	<code>-DAYS_BIRTH / 365</code>	More interpretable than days; banks care about applicant age

YEARS_EMPLOYED	-DAYS_EMPLOYED / 365 (0 if unemployed)	Employment stability is a strong credit predictor
IS_UNEMPLOYED	1 if DAYS_EMPLOYED == 365243 else 0	Binary flag for unemployment (sentinel value 365243)
INCOME_PER_MEMBER	AMT_INCOME_TOTAL / CNT_FAM_MEMBERS	Income adequacy relative to household size

Preprocessing Pipeline

- **Categorical encoding:** LabelEncoder applied to 5 categorical columns — saved as encoder.pkl
- **Numerical scaling:** StandardScaler applied to 14 numerical columns — saved as scaler.pkl
- **Feature order:** Categorical features first, then numerical — must match exactly at inference time

Milestone 4: Model Training & Comparison

Class Imbalance Handling

The dataset has an 87%/13% class imbalance (Approved/Rejected). All models use balanced weights:

- **Scikit-learn models:** `class_weight='balanced'`
- **XGBoost:** `scale_pos_weight = neg_count / pos_count`

Model Training Results

Model	Accuracy	ROC-AUC	Training Notes
Logistic Regression	74.3%	0.81	L2 regularization, max_iter=1000
Decision Tree	76.8%	0.77	Default depth; prone to overfitting
Random Forest	81.91%	0.89	n_estimators=100, best overall model
XGBoost	79.5%	0.87	scale_pos_weight=6.7 (neg/pos ratio)

Best Model: Random Forest (highest accuracy + ROC-AUC)

All model metrics are saved in `models/metadata.pkl` and displayed on the dashboard.

Milestone 5: Flask Web Application Development

Application Routes

Route	Method	Description
/	GET	Dashboard — model comparison table + scenario cards
/predict	GET	Blank prediction form
/predict	POST	Form submission → preprocessing → ML inference → result
/history	GET	Session history of last 10 predictions
/clear_history	POST	Clear session history

Preprocessing at Inference (`preprocess_input()`)

- 1 Extract 5 categorical values from form → `LabelEncoder.transform()` → array
- 2 Extract 14 numerical values from form → `StandardScaler.transform()` → scaled array
- 3 Concatenate: `[cat_encoded] + [num_scaled]` → 19-element feature vector
- 4 Reshape to (1, 19) → `model.predict()` + `model.predict_proba()`

Frontend Architecture

- **base.html:** Shared layout with Bootstrap 5.3 CDN, Bootstrap Icons, navbar
- **index.html:** Hero section, model comparison table (from metadata.pkl), scenario cards with JS pre-fill
- **predict.html:** 4-section form (Personal, Financial, Employment, Education & Housing), result card with animated confidence bar
- **history.html:** Summary statistics (approved_count via Jinja2 selectattr), collapsible prediction detail cards
- **style.css:** CSS variables --primary-blue: #1a3a6b, --accent-gold: #c8a84b, fadeInUp animation for result card
- **script.js:** SCENARIOS object, fillScenario(), updateIncomeDisplay(), updateUnemployed(), animateProgressBars()

Milestone 6: Deployment

Render Deployment Configuration

Procfile:

```
web: gunicorn app:app
```

render.yaml:

```
services:
- type: web
  name: credit-card-approval-prediction
  runtime: python
  buildCommand: pip install -r requirements.txt
  startCommand: gunicorn app:app
```

Deployment Steps

- 1 Committed all source code + models/*.pkl artifacts to GitHub master branch
- 2 Connected GitHub repository to Render via Render dashboard
- 3 Render automatically installs dependencies and starts Gunicorn on each push to master
- 4 Application served at: <https://credit-card-approval-prediction-ra22.onrender.com>

CI/CD Pipeline

Every `git push origin master` triggers:

- 1 Render detects new commit
- 2 Pulls latest code
- 3 Runs `pip install -r requirements.txt`
- 4 Starts `gunicorn app:app`
- 5 New deployment live in ~2 minutes